

Mini Lesson Sample Script

1. Multiply a positive number by a positive number and a positive number by a negative number to review known sign rules.

Evaluate each product.

a) $6 \cdot 2$

b) $6 \cdot (-2)$

Answer: a) 12; b) -12

T: Let's warm up. What's $6 \cdot 2$?

S: 12.

T: And how do you know the product is positive?

S: A positive times a positive is positive.

T: Now try $6 \cdot (-2)$. You can think of this as 6 groups of -2.

S: 6 groups of -2 is -12.

T: What sign rule does that show?

S: A positive times a negative is negative.

2. Use the distributive property to discover that the product of two negative numbers is positive.

Evaluate the expression $-4 \cdot (7 + (-2))$ two ways.

First, evaluate what is inside the parentheses, then multiply.

T: We're going to evaluate this expression two different ways. The answers should match. Let's start by simplifying inside the parentheses. What is $7 + (-2)$?

S: 5.

Then, use the distributive property to rewrite the expression as a sum of two products. Use what you know about the first product and the total to figure out the value of the second product.

Answer: $-4 \cdot (7 + (-2)) = -4 \cdot 5 = -20$; using the distributive property, $-4 \cdot 7 + -4 \cdot (-2) = -28 + -4 \cdot (-2) = -20$, so $-4 \cdot (-2) = 8$.

T: So now we have $-4 \cdot 5$. What's that?

S: -20.

T: Good. So the whole expression equals -20. Now let's try it again, this time using the distributive property. Distribute the -4 to the 7 and to the -2. What do you get?

S: $-4 \cdot 7$ plus $-4 \cdot (-2)$.

T: What's $-4 \cdot 7$?

S: -28.

T: So we know -28 plus $-4 \cdot (-2)$ has to equal -20, because both methods give the same answer. What number do we add to -28 to get -20?

S: 8.

T: So $-4 \cdot (-2)$ has to equal 8. That's a negative times a negative, and the product is positive. For the distributive property to work for all numbers, a negative times a negative has to be positive.

3. Apply the rules for multiplying signed numbers to evaluate products involving negative factors.

Evaluate each product.

a) $-5 \cdot (-3)$

b) $-7 \cdot (-4)$

c) $-6 \cdot 4$

Answer: a) 15; b) 28; c) -24

T: Try these on your own. Name the sign rule you use for each.

S: (Work.)

T: What did you get for $-5 \cdot (-3)$?

S: 15. Negative times a negative is positive.

T: And $-7 \cdot (-4)$?

S: 28. Same rule.

T: How about $-6 \cdot 4$?

S: -24. A negative times a positive is negative.